FastSpeech 2 for English and Telugu

Technical Report: Initial Evaluation Assignment, Speech Processing Lab, IIIT Hyderabad

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1 Summary

I trained two FastSpeech 2 text-to-speech models from scratch: one for **English** (Svarah dataset, many speakers, Indian-accented speech) and one for **Telugu** (IndicTTS Telugu, one male voice). Each model used about 5 hours of selected training speech (about 4.7 h for English and 4.4 h for Telugu after silence was trimmed) and was trained for 50,000 steps. The code is a modernised version of the reference implementation (ming024/FastSpeech2). Phone durations come from a forced aligner, and the pre-trained HiFi-GAN universal vocoder converts the predicted mel-spectrograms to audio.

To measure intelligibility, I synthesised 80 held-out sentences per language and transcribed them with speech recognisers. The word error rate is 11.7% for English (6.3% on real recordings) and 6.9% for Telugu (5.2% on real recordings). A native Telugu speaker listened to the five Telugu test sentences and confirmed that every word is pronounced as written. Both voices sound synthetic, and the Telugu voice sounds clearly robotic. Validation loss stops improving early because the data is small, but the final checkpoints are still the most intelligible, so they are the ones submitted. The main failures are rare or foreign words in English ("antidisestablishmentarianism", the Telugu name "andaroo"). All code, checkpoints, audio and plots are in the repository (Section 9).

2 Implementation approach

Architecture. FastSpeech 2 generates all mel frames in parallel. It has a phoneme encoder (4 feed-forward Transformer blocks, hidden size 256, 2 attention heads, convolutional feed-forward layers with kernel sizes 9 and 1), a *variance adaptor*, a mel decoder (6 blocks) and a 5-layer convolutional post-net. The variance adaptor has three predictors (duration, pitch, energy; each two 1-D convolutions with 256 channels, kernel size 3, dropout 0.5) and a length regulator that repeats each phoneme encoding for its duration. Pitch and energy are predicted once per phoneme, quantised into 256 bins and added to the encoder output as learned embeddings. During training the decoder receives the *true* duration, pitch and energy, and the predictors learn to match them (mean squared error, duration on $\log(d+1)$). At inference the predicted values are used instead; these predictions are what the variance-adaptor plots show. The loss is the mel L1 error (before and after the post-net) plus the three predictor errors. Each model has about 34.6 M parameters.

Changes to the reference code. (i) The code uses current libraries (PyTorch 2 attention, librosa 0.10), a vectorised length regulator, fp16 mixed precision, resumable training, best-checkpoint tracking and small fp16 inference checkpoints, in one package (`fs2/`) shared by both languages. (ii) Features are read directly from Montreal Forced Aligner (MFA) 3 TextGrids. MFA 3 marks silence with an empty label, so leading and trailing silence is trimmed and silence inside a sentence becomes the symbol `sp`. Pitch and energy statistics are computed on training data only. (iii) Batches group utterances of similar length, which reduces padding and roughly halves the training time.

English. The model is multi-speaker: a learned speaker embedding (114 pseudo-speakers, defined in Section 3) is added to the encoder output. The 70 symbols are 45 vowel symbols (15 ARPAbet vowels $\times$ 3 stress levels), 24 consonants and `sp`. The text front-end expands numbers, times and years into words, then looks up pronunciations with CMUdict and `g2p_en` (a neural model for unknown words). Pronunciations are produced for whole phrases, not single words, so that the part-of-speech tagger can choose the right pronunciation of words such as "conduct" (noun or verb). AM/PM is spelled out, and punctuation becomes `sp`. Alignments come from MFA with the pretrained `english_us_arpa` model and an extended dictionary: the MFA dictionary plus MFA G2P pronunciations for 1,173 corpus words (4.23% of all word tokens) that the dictionary lacks.

Telugu. No pretrained Telugu aligner or pronunciation dictionary was available, and I did not write phoneme rules by hand. Instead, the text is converted letter by letter with the ISO 15919 romanisation standard (using Aksharamukha), which maps each Telugu letter to exactly one symbol and can be reversed. It marks long vowels (ā ī ū ē ō), retroflex consonants (ṭ ḍ ṇ ḷ), the anusvara (ṁ), and writes double consonants by repeating the letter. Aspirated consonants (kh, gh, ch, jh, ṭh, ḍh, th, dh, ph, bh) are single symbols. This gives 49 letter symbols plus `sp` (50 in total). Punctuation becomes `sp`; word boundaries are not marked, as in the English pipeline. Text is first normalised (Unicode NFC, zero-

width characters removed) and digits are expanded into Telugu number words with num2words, with two small fixes. Since the symbols are the letters themselves, the aligner dictionary simply maps each word to its symbols, and MFA was *trained from scratch* on the 2,830 selected clips in single-speaker mode. The first pass aligned 2,530 clips. Re-aligning with the trained model and a wider search beam (200, retry 800) recovered the other 300, so no clip was lost. The number-reading rules (for example, 10:45 read as two numbers) are my own choices. I checked the Telugu output with a Telugu speech recogniser and with one native speaker (Section 6).

3 Data selection, split and preprocessing

All audio was converted to mono, 22,050 Hz, 16-bit WAV, the sampling rate of the vocoder. Selection used fixed rules (not listening), with random seed 42.

	English (Svarah)		Telugu (IndicTTS)	
	Train	Validation	Train	Validation
Number of clips	3,183	700	2,361	469
Hours selected (before trimming)	5.07	1.04	5.10	1.00
Hours after trimming silence	4.71	0.96	4.35	0.85
Number of speakers	114	113	1	1
Sentences shared with the other split	0	0	0	0

Table 1: Data used for each model. The assignment asks for about 5 h of training and 1 h of validation speech. Trimming leading and trailing silence reduces the hours the model actually sees.

English selection. Svarah has 117 speakers but no speaker ID, so I defined a *pseudo-speaker* as each unique combination of eight speaker-metadata fields. This gives 115 combinations, so a few real speakers may be merged, and 114 of them appear in the selected data. A clip was kept if it is 1.5–15 s long; its text has only letters, apostrophes and basic punctuation (no digits) and at least 3 words; it has 5–22 characters per second (this removes badly matched transcripts); its peak level is at least 0.03; and at most 0.5% of its samples are clipped. This left a pool of 6.41 h. From it, the clips with the worst signal-to-noise estimate (ratio of the 90th to the 10th percentile of frame energy) were dropped until 6.1 h remained, then the data was split into 5.07 h and 1.04 h.

Telugu selection. I used only the male voice, so there is one consistent speaker. A clip was kept if it is 1.5–15 s long; its text has no digits and no rare letter (ఱ), only Telugu script and basic punctuation, 8–16.5 characters per second, and every letter maps in the front-end; and the same level and clipping checks pass. The 3,648 clips that passed were taken in random order until 6.1 h was reached (2,830 clips).

Split. The split is at utterance level and balanced by speaker, and *all clips with the same sentence go to the same side* (Svarah repeats sentences across speakers). No more than 40% of one speaker's time goes to validation. So no sentence appears in both training and validation. I did not split by speaker for English, because a speaker-embedding table cannot represent a speaker it has never seen.

Features. Mel-spectrogram: STFT 1024/256/1024, 80 mel bands, 0–8,000 Hz, natural-log magnitude (matching HiFi-GAN). Pitch (F0): pyworld (DIO with StoneMask refinement; search range 60–800 Hz for English, 50–800 Hz for Telugu), filled in over unvoiced frames and averaged per phoneme. Energy: L2 norm of the STFT magnitude, averaged per phoneme. Both are normalised with training-set statistics (English F0 mean 174.6 Hz, standard deviation 55.6; Telugu F0 mean 187.1 Hz, standard deviation 42.2).

4 Training configuration

Optimiser	Adam (β = 0.9, 0.98; $\epsilon = 10^{-9}$), gradient clipping 1.0, no weight decay
Learning rate	Noam warm-up over 3,000 steps (peak about $1.1 \cdot 10^{-3}$), then multiplied by 0.3 at steps 35,000 and 45,000
Batch and length	32 utterances per batch, 50,000 steps (about 500 epochs for English, 680 for Telugu)
Precision, hardware	fp16 mixed precision, one Tesla T4 GPU per language, both trained at the same time
Training time	English about 3.5 h (4.1 steps/s); Telugu about 4.3 h (3.3 steps/s)
Validation	every 1,000 steps on the full validation set (no dropout)
Vocoder	pre-trained HiFi-GAN universal (not trained here)
Checkpoints saved	best validation loss and final; the final one (step 50,000) is submitted

Table 2: Training settings (identical for both languages except the speaker table and the F0 range).

This is far less training than the reference recipe (900,000 steps at batch size 16), so the models may be under-trained and over-smooth.

5 Training and validation loss

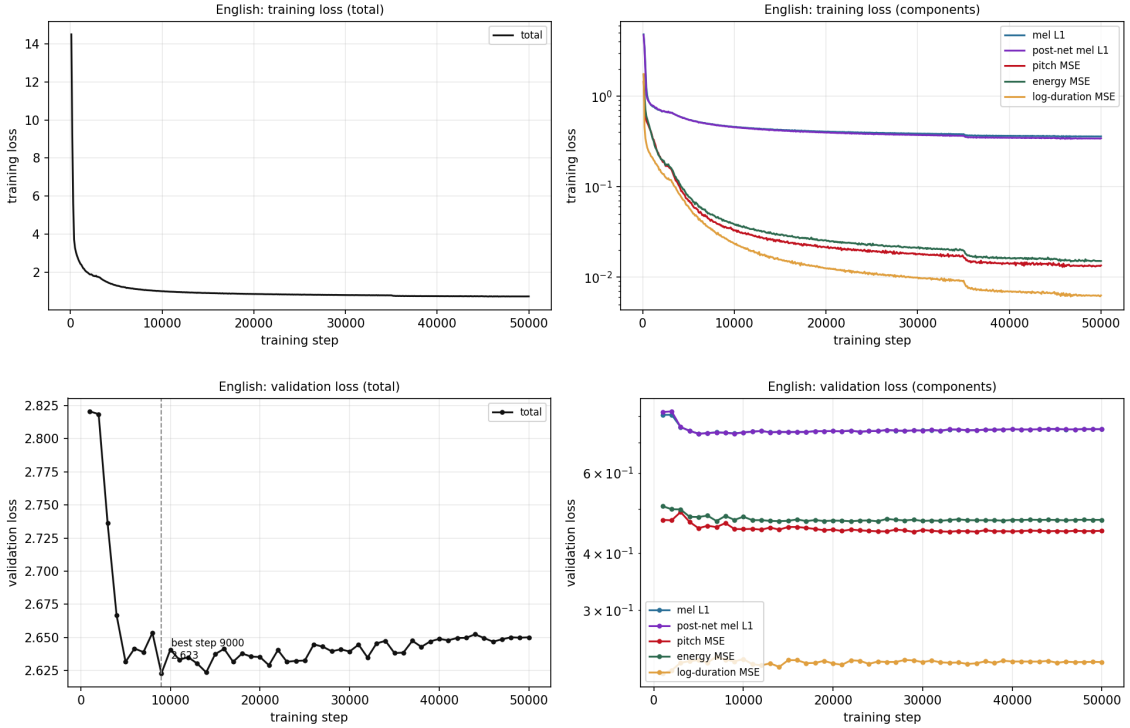


Figure 1: English training loss (top) and validation loss (bottom). Left: total loss. Right: components on a log axis (mel L1 before and after the post-net, pitch, energy, log-duration). The best validation loss is marked.

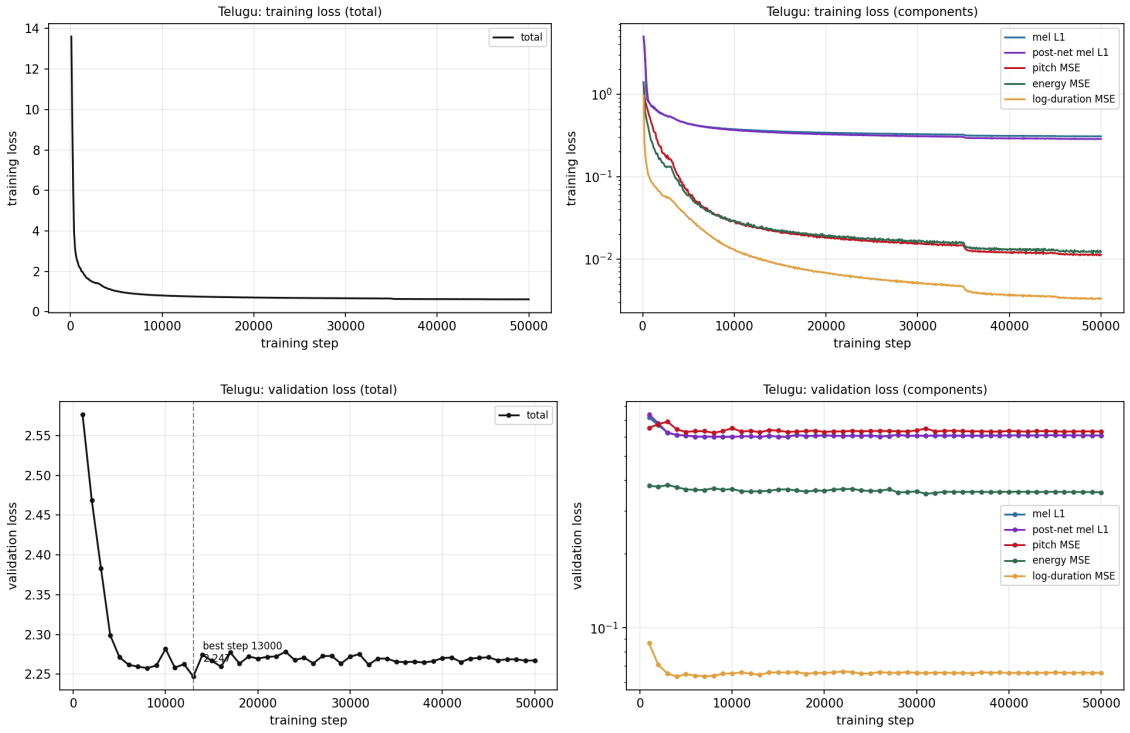


Figure 2: Telugu training loss (top) and validation loss (bottom), same layout.

Convergence. Training loss falls quickly: from about 14 at step 100 to about 3 at step 500 and to about 1 by step 9,000 (English 1.04, Telugu 0.83), then improves slowly. The small drops after steps 35,000 and 45,000 come from the learning-rate cuts. There were no loss spikes or divergence.

Overfitting. Validation loss is lowest early (English step 9,000: 2.623; Telugu step 13,000: 2.247) and then stays flat, ending slightly higher (English +1.0%, Telugu +0.9% at step 50,000), while training loss keeps falling. The final gap is large (0.74 against 2.65 for English; 0.62 against 2.27 for Telugu). Most of the gap is in the pitch and energy terms. In training, the true pitch and energy are fed to the decoder, so the model can memorise each training utterance, and these errors fall to about 0.01. On unseen sentences they stay near 0.45 and 0.47 (English) and 0.63 and 0.36 (Telugu). Part

		Total	Mel	Post-net	Pitch	Energy	Duration
English	Train, step 50k	0.737	0.360	0.342	0.013	0.015	0.006
	Validation, best (step 9k)	2.623	0.733	0.733	0.452	0.473	0.231
	Validation, step 50k	2.650	0.749	0.749	0.448	0.473	0.231
Telugu	Train, step 50k	0.619	0.307	0.285	0.011	0.012	0.003
	Validation, best (step 13k)	2.247	0.599	0.599	0.625	0.360	0.064
	Validation, step 50k	2.267	0.608	0.608	0.630	0.356	0.066

Table 3: Loss values. Training values are measured with dropout on, validation values with dropout off.

of this floor is unavoidable: how high or loud a speaker says a phoneme in a new sentence cannot be fully predicted from text, so the predictors output an average. Telugu has a much lower duration error than English (0.066 against 0.231) because it has a single, consistent voice, but a higher pitch error. The mel error on validation (0.75 and 0.61) is about twice the training value, which is a normal generalisation gap for small data.

Choice of checkpoint. Validation loss turned out to be a poor guide. In both languages the final checkpoint is *more* intelligible than the best-validation one (English character/word error 5.68/11.67% against 6.45/11.98%; Telugu 4.21/6.92% against 4.63/7.05%; Table 4). I therefore submit the final (step 50,000) checkpoints. Outputs of the best-validation checkpoints are kept in `Results/*/extra_validation_best_checkpoint`. Possible remedies for the overfitting are stronger regularisation, data augmentation, more data, or a pretrained multilingual encoder.

6 Generated speech

Objective check. I synthesised 80 held-out validation sentences per language (English in the voice of each sentence's own pseudo-speaker) and transcribed them with Whisper-small (English) and a Telugu Whisper large-v2 model (vasista22/whisper-telugu-large-v2). Transcripts were compared with the input text, ignoring case and punctuation. The same recogniser on the real recordings gives the best result that can be expected. Both models are understood almost

	English (Whisper-small)		Telugu (Whisper Telugu large-v2)	
	CER %	WER %	CER %	WER %
Real recordings	3.15	6.25	2.72	5.15
Model, final (step 50k)	5.68	11.67	4.21	6.92
Model, best validation	6.45	11.98	4.63	7.05

Table 4: Intelligibility on 80 held-out sentences per language (CER = character error rate, WER = word error rate; lower is better).

as well as real speech. This test measures intelligibility only: recognisers have strong language models and can correct small errors, and the test says nothing about naturalness.

Native-speaker check (Telugu). One native Telugu speaker listened to T1–T4. Every word was pronounced as written, and the voice sounds robotic. This was an informal check by a single listener, not a formal listening test, and the speaker was not asked specifically whether numbers such as "2026" and "10:45" are read the usual way.

Observations per test sentence (final checkpoint; "transcript" is what the recogniser heard; English uses the most frequent training pseudo-speaker).

- **E1** ("The antidisestablishmentarianism movement…", 5.20 s). Transcript: "The entire distrust of this materialism movement in considerable momentum during the parliamentary proceedings." The 27-phone long word takes only 1.7 s (62 ms per phone on average, shortest 35 ms), so its syllables are blurred; the rest of the sentence is correct. The model fails on this very long, rare word.
- **E2a** (noun "conduct", 3.52 s). The transcript matches the text exactly. The final sound is lengthened (221 ms, against a median phone length of 58 ms).
- **E2b** (verb "conduct", "The researchers will conduct an experiment…"). In the first synthesis the front-end gave "conduct" the noun stress pattern (K AA1 N D AH0 K T); the recogniser heard "Researchers will conduct an experiment to evaluate the proposed method" (it missed "The"). The front-end was then changed to look at the whole phrase, and it now gives the verb pattern (K AH0 N D AH1 K T). The sentence was regenerated with the corrected front-end (see `Results/English`) and was not re-scored by the recogniser.
- **E3** ("The rhyth m of the orchestra…", typing error kept as given, 2.61 s). The front-end reads "rhyth" as /raɪð/ and "m" as the letter name "em", which adds a stressed vowel (2.61 s against 2.48 s for the corrected sentence). The recogniser still heard "The rhythm of the orchestra changed unexpectedly", because its language model repairs the input. So the transcript makes the audio look better than it is. The corrected sentence (extra file) gives the same transcript.
- **E4** ("The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.", 6.41 s). Transcript: "The meeting will start at 10.45 am on 12 November 2026, so on the rush to drive on time." The time and date are

read correctly (expanded to words, AM spelled out). The comma gives a 569 ms pause. “andaroo” is a Telugu word that the English G2P cannot pronounce correctly, and it is not recognised.

- **T1** (long loanword “న్యూరోఫార్మకోలాజికల్”, 4.46 s). The transcript equals the input words, and the native speaker found all words correct. The 20-symbol loanword takes 1.46 s.
- **T2a, T2b** (two different sentences that contain the word “పట్టు”: “grip” in T2a, “silk” in T2b; 2.15 s and 2.35 s). Both transcripts match exactly. Both words use the same symbols (p a ṭ ṭ u). The two ṭ symbols last 70 and 46 ms, 116 ms together, about twice a single consonant.
- **T3** (“పులి అడవిలో నిశ్శబ్దంగా నడిచింది”, 1.92 s). The recogniser heard “పులి అడవిలోని శబ్దంగా నడిచింది”, which drops one “ni” and merges the words. The model does produce a long “śś” (174 ms), and the native speaker found every word correct, so this looks like a recogniser mistake rather than a synthesis error.
- **T4** (“…2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి…”, 6.93 s). Numbers are read as words (“రెండు వేల ఇరవై ఆరు”, “పన్నెండు”, “పది నలభై ఐదు”). The comma gives a 151 ms pause, preceded by a lengthened final vowel (163 ms). This is the fastest sentence (17.7 phones per second) because 124 symbols fit in 6.9 s.

7 Variance-adaptor analysis

All 30 plots (duration, pitch and energy for each of the test sentences) are in `Results/<Language>/variance_adaptor_plots/`. Table 5 summarises them, and Figures 3 and 4 show the two longest sentences. Pitch and energy are one value per phoneme, drawn as steps as wide as the predicted duration. Pitch is in Hz and energy is the L2 norm of the STFT magnitude. Durations are whole numbers of 11.6 ms frames, so the bars look quantised.

ID	Length (s)	Symbols	Phones/s	F0 mean / sd	F0 range	F0 first → last fifth	Pause (ms)
E1	5.20	87	16.7	131 / 10	109–160	137→121	–
E2a	3.52	51	14.5	131 / 12	106–159	146→117	–
E2b	3.55	56	15.8	129 / 14	109–168	145→121	–
E3	2.61	37	14.2	133 / 14	116–176	146→123	–
E3_corrected_extra	2.48	37	14.9	134 / 12	118–166	145→123	–
E4	6.41	81	12.5	140 / 19	90–198	161→121	569
T1	4.46	67	15.0	190 / 34	126–269	226→144	–
T2a	2.15	32	14.9	194 / 34	115–256	225→156	–
T2b	2.35	37	15.7	183 / 35	104–252	164→138	–
T3	1.92	31	16.1	183 / 32	115–235	204→136	–
T4	6.93	124	17.7	177 / 23	110–228	189→157	151

Table 5: Summary of the predicted values. F0 is in Hz and `sp` is excluded from the F0 statistics. “First → last fifth” compares the mean F0 of the first and last fifth of the phonemes.

Duration. The median phone length is 58 ms in both languages (mean 67 ms for English, 61 ms for Telugu), and speaking rate is 12.5–17.7 phones per second. In English, phonemes with primary stress last longer than unstressed ones (69 ms against 59 ms on average over the test sentences), and the last sound of a sentence is stretched (E3: final vowel 209 ms; E2a: final /s/ 221 ms). This shows that the model has learned phrase-final lengthening. The only pause inside a sentence is at the comma (E4: 569 ms; T4: 151 ms). In Telugu the predicted durations follow the length contrast of the language: long vowels average 91 ms against 53 ms for short vowels, and double consonants are about twice as long as single ones (ṭṭ 116 ms, śś 174 ms, cc 128 ms).

Pitch. Every sentence has a falling pitch: the first fifth of the phonemes has a higher mean F0 than the last fifth (E1: 137 → 121 Hz; T1: 226 → 144 Hz). English contours are narrow (standard deviation 10–19 Hz, mean F0 about 130–140 Hz for the chosen low-pitched speaker) and Telugu contours are wider (23–35 Hz, mean about 180–195 Hz). Each phoneme gets one fixed value with no randomness, so contours are smooth steps without the small variations of real speech. This is the main reason for the synthetic, robotic sound. The Telugu F0 mean (≈190 Hz) is high for a male voice. It may be a high-pitched speaker or errors in pitch tracking (octave errors); I did not investigate this.

Energy. Energy is high on vowels and low on consonants (English mean 77 against 39; Telugu 50 against 28), highest on stressed vowels (up to 121 in E2a), near zero in pauses, and it falls at the end of the sentence (T1: 57 → 21), which is the usual drop at the end of a statement.

Context effects. The same word gets different values in different sentences. For example, “పట్టు” has the same symbols in T2a and T2b, but its phone durations and F0 differ slightly, so the predictors use the sentence context. In E4, the 569 ms pause is followed by a reset: F0 goes from 82 Hz on `sp` to 128 Hz on the next sound.

8 English and Telugu compared, limitations and improvements

Comparison. Telugu has lower error rates than English, both in absolute terms and compared with real recordings (character error 1.5 points above real speech against 2.5 points; word error 1.8 against 5.4 points). Likely reasons: Telugu

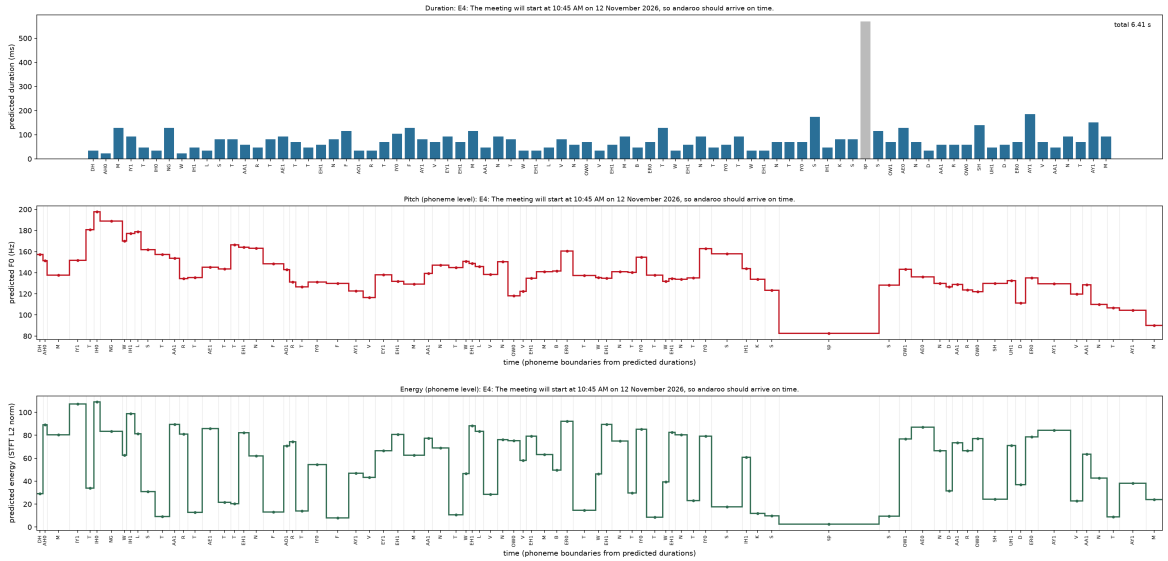


Figure 3: E4: predicted duration, pitch and energy. The grey bar in the duration plot is the 569 ms pause at the comma.

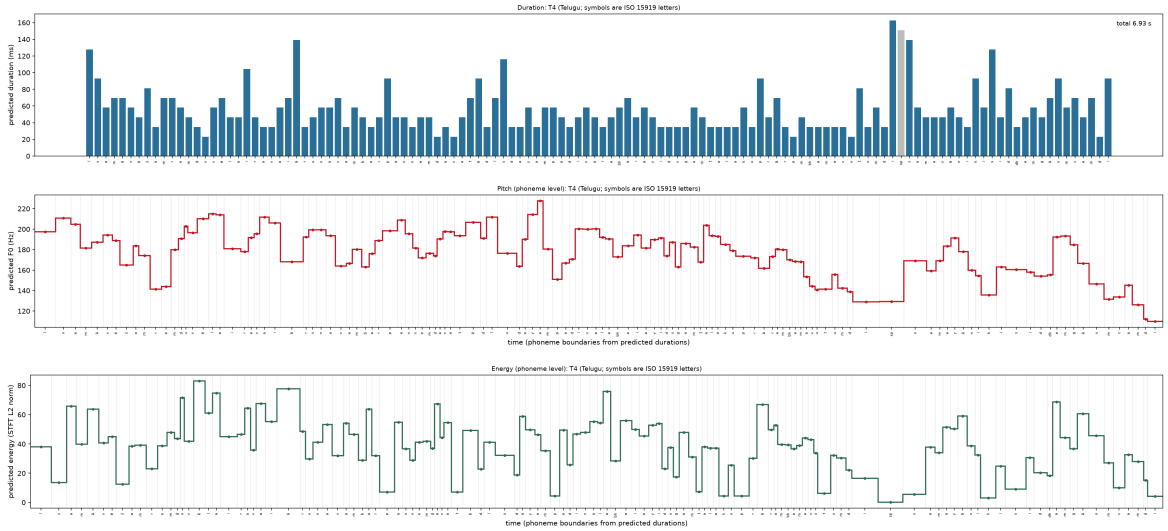


Figure 4: T4 (124 symbols; ISO 15919 letters on the axis): predicted duration, pitch and energy.

has one speaker; the IndicTTS recordings are clean studio recordings, while Svarah has varied microphones and accents; the Telugu writing system is almost phonetic, so the symbol conversion is exact, while English needs a dictionary and a guessing G2P; and the Telugu sentences come from a narrower domain. English must cover 114 pseudo-speakers with about 4.7 h, which is only about 2.5 minutes per voice. Telugu has a higher pitch error and wider F0 range; English has a higher duration error.

Failures and limitations. (1) Rare and foreign words fail in English (“antidisestablishmentarianism” in E1, “andaroo” in E4): the front-end has to guess, and the model squeezes long unfamiliar words into very short phones. The Telugu loanword in T1 works because the script spells it out. (2) The typing error in E3 shows that the front-end cannot repair bad input; the recogniser’s language model hides this. (3) Naturalness is limited. Both voices are over-smooth, and the Telugu one is clearly robotic: the predictors have no randomness, there is little data, and the vocoder was not trained on these voices. (4) The variance predictors overfit. (5) The number-reading rules are my own and were checked only with the recogniser and one speaker. (6) The evidence is limited: one recogniser per language and one native listener, no formal listening test (MOS). English pseudo-speakers may merge a few real speakers, and the two recognisers differ, so the English and Telugu error rates are not strictly comparable. (7) The English E2b homograph was corrected after the first synthesis (see E2b), and that sentence was not re-scored by the recogniser.

Possible improvements. More data and a pretrained or multilingual encoder; variance models that sample instead of predicting one value (for example flow or diffusion based) to reduce the monotone sound; a vocoder fine-tuned on these voices; a better English G2P with a strategy for rare words and foreign names; speaker-balanced sampling; and for Telugu, a number and abbreviation grammar checked by native speakers and a formal listening test.

9 Reproducibility

Software: Python 3, PyTorch 2, librosa 0.10, pyworld, g2p_en (with NLTK data), num2words, aksharamukha, transformers (only for the recogniser check) and Montreal Forced Aligner 3.4.3 (see `requirements.txt`). Training used Kaggle with two Tesla T4 GPUs. The two configuration files (`English/config.yaml`, `Telugu/config.yaml`) differ only in paths and the F0 floor. The code is organised so that `English/` and `Telugu/` are the only entry points (`preprocess.py`, `train.py`, `inference.py`, one `config.yaml` and one `frontend.py` each). The language-independent model code is shared in `fs2/`, and `scripts/` holds data selection, the aligner scripts, evaluation and plotting. Commands, run from the repository root:

```
python scripts/prepare_data.py ; python scripts/build_dictionaries.py ; bash scripts/align.sh
python English/preprocess.py ; python Telugu/preprocess.py
CUDA_VISIBLE_DEVICES=0 python English/train.py
CUDA_VISIBLE_DEVICES=1 python Telugu/train.py
python English/inference.py   # uses English/checkpoints/model_final_fp16.pt, writes to Results/English
python Telugu/inference.py    # uses Telugu/checkpoints/model_final_fp16.pt, writes to Results/Telugu
```

Data selection, split and export are in `scripts/prepare_data.py`. The `scripts/build_dictionaries.py` and `scripts/align.sh` scripts were rewritten from the notebook cells I used on Kaggle. The HiFi-GAN weights are downloaded from the reference repository on first use. The final checkpoints (step 50,000, about 69 MB each) belong in `English/checkpoints` and `Telugu/checkpoints`.

References. Y. Ren et al., *FastSpeech 2: Fast and High-Quality End-to-End Text to Speech*, ICLR 2021 (arXiv:2006.04558). ming024/FastSpeech2 (reference implementation). J. Kong et al., *HiFi-GAN*, NeurIPS 2020. M. McAuliffe et al., *Montreal Forced Aligner*, Interspeech 2017. Datasets: Svarah (AI4Bharat), IndicTTS (IIT Madras). Recognisers: Whisper (OpenAI), vasista22/whisper-telugu-large-v2.